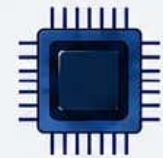
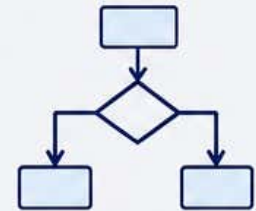




1010101010
0101010101
1010101010
0101010101



Syllabus



Minor Course: Laboratory on Software Engineering and Java Programming

Level	Semester	Course Code	Course Name	Credits	Teaching Hours	Exam Duration	Max Marks
5.5	VI	112268	Laboratory on Software Engineering and Java Programming	2	60	4Hrs.	50

Course Objectives:	1. To understand the basic concepts of Java programming language. 2. To develop problem-solving skills using Java programs. 3. To provide the knowledge of basic of Computer Network. 4. To understand importance of OSI model in Computer Networking.						
Course Outcomes:	On completion of the following syllabus the students will be able to - 1. Understand and apply the fundamentals of Java programming language. 2. Execute Write, compile, debug, and execute Java programs successfully. 3. Use decision-making statements, loops, and functions to solve programming problems. 4. Provide the knowledge of basic of Computer Network. 5. Understand importance of OSI model in Computer Networking.						
Contents					Workload Allotted	Weightage of Marks Allotted	Incorporation of Pedagogies
	List of Practical: 1. Write a program in Java to Swap Two Numbers using third Variables. 2. Write a program in Java to determine whether an input number is an even number. 3. Write a program in Java to Find Greatest of Three Number 4. Write a program in Java to Switch Case Statement 5. Write a program in Java to Print Factorial of number 6. Write a program in Java to Print Prime Number 7. Write a program in Java to Generate Fibonacci Series 8. Write a Java Program for Printing Multiplication table using do-while statement. 9. Write a Program in java by using while loop to reverse the inputted number 10. Write a program in Java to Print following pattern * * *						Use any pedagogical technique suitable for the topic

* * *
* * * *
* * * * *

11. Write a program in Java to print the following pattern:

1
12
123
1234
12345

12. Write a program in Java code to print the following pattern:

1
22
333
4444
55555

13. Write a program in Java code to print the following pattern:

1
2 3
4 5 6
7 8 9 10

14. Write a program in Java to demonstrate classes and objects.

15. Write a program in Java to demonstrate method overloading

16. Write a program in Java to demonstrate constructor.

17. Write a program in Java to demonstrate constructor overloading.

18. Write a program in Java to demonstrate single inheritance

19. Study of various phases of SDLC.

20. Identifying the Requirements from Problem Statements.

21. Modeling UML Use Case Diagrams and Capturing Use Case Scenarios.

22. Modeling UML ER Diagrams and Sequence diagrams.

23. State chart and Activity Modeling.

24. Modeling Data Flow Diagram.

25. Study of various testing methods

References:	Weblink to Equivalent MOOC on SWAYAM if relevant: 1. https://onlinecourses.swayam2.ac.in/e-learning/preview/aic20_sp13 2. https://onlinecourses.nptel.ac.in/e-learning/preview/noc25_cs108 3. https://cse.iitk.ac.in/pages/CS455.html 4. https://se-iitkgp.vlabs.ac.in/			
--------------------	---	--	--	--

Assessment Rubric: External Practical Course

SANT GADGE BABA AMRAVATI UNIVERSITY, AMRAVATI PRACTICAL EXAMINATION B.Sc. III, SEMESTER – VI (NEP)				
Practical	Course Code: 112268	Laboratory on Software Engineering and Java Programming	Max Marks: 25	Time: 4 Hrs.
Sr. No.	Exercise			Marks
1.	Single Program Writing, Algorithm, Flowchart, Execution, with Sample inputs and Outputs, and Printout			10
2.	Submitted Record book (Preferably handwritten)			05
3.	External Viva Voce			10

Assessment Rubric: Internal Practical Course

SANT GADGE BABA AMRAVATI UNIVERSITY, AMRAVATI PRACTICAL EXAMINATION B.Sc. III, SEMESTER – VI (NEP)			
Practical	Course Code: 112268	Laboratory on Software Engineering and Java Programming	Max Marks: 25
Sr. No.	Assessment Criteria		Marks
1.	Attendance/Regularity in practical's / Students' overall performance		05
2.	Record book Preparation (Preferably handwritten)		10
3.	Internal Viva Voce		10